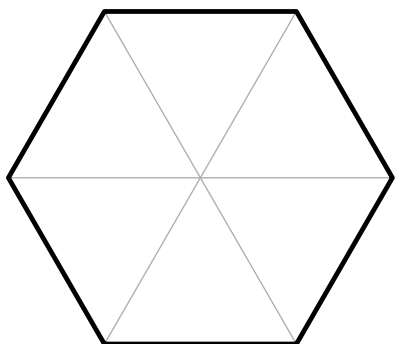


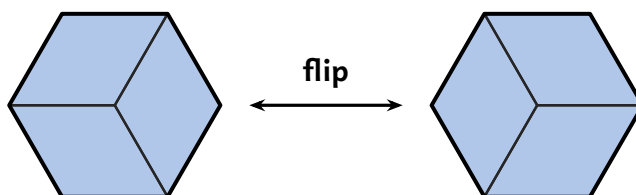
1 Three blues, two ways

Name: _____

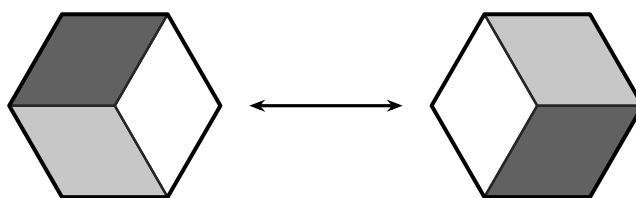
Today we only use **blue rhombi**. On our grid a blue can sit in three directions:



1. Cover this hexagon with 3 blues. There are exactly **two** ways. Build both.
2. Lift the three blues and turn them, so that one way becomes the other. This move is called a **flip**.

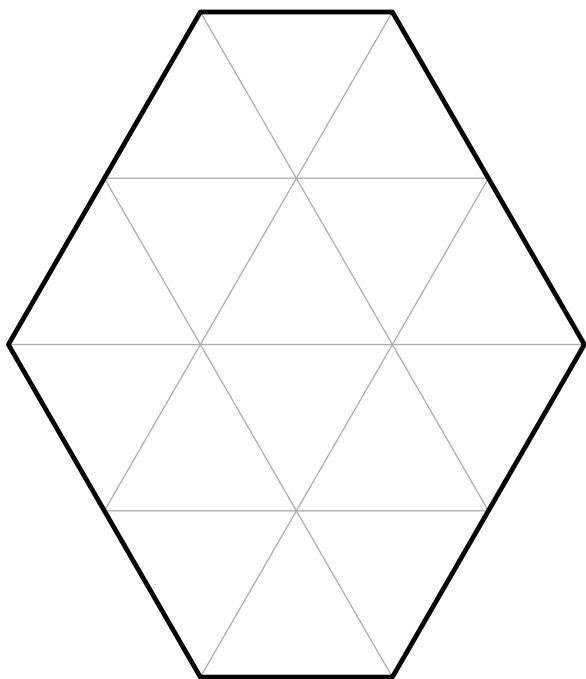


3. Look in 3D. Here are the same two tilings, shaded: one direction white, one light gray, one dark gray. Stare at them. What do you see? What does a flip do to the picture?



Coming up. Bigger hexagons have many tilings. You will hunt for all of them, connect them by flips, and find a secret code hiding inside every tiling.

2 The tall hexagon

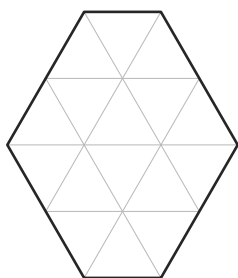


Cover this board with 8 blue rhombi. This is called a **tiling**.

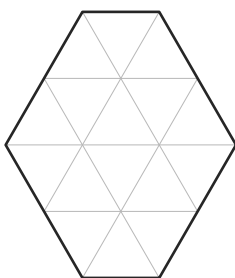
How many *different* tilings can your group find? Draw each new one in a box below and give it a letter.

Tips.

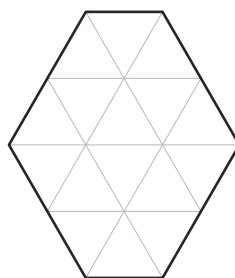
- One person builds, one checks that it is new, one draws.
- Stuck? Look for three blues that make a small hexagon, and flip them!



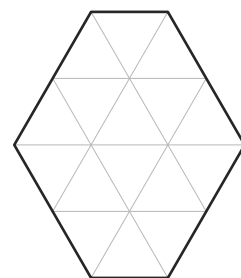
A



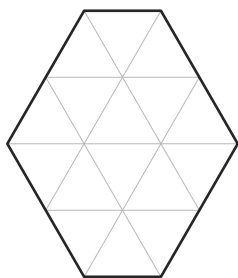
B



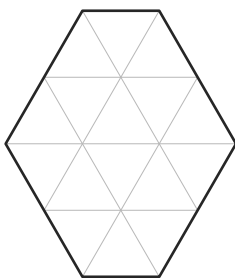
C



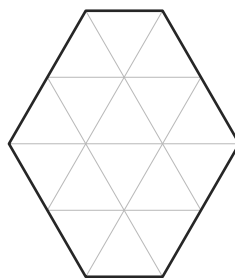
D



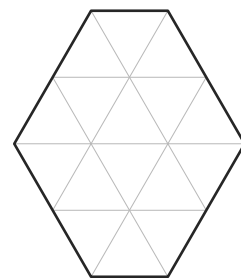
E



F



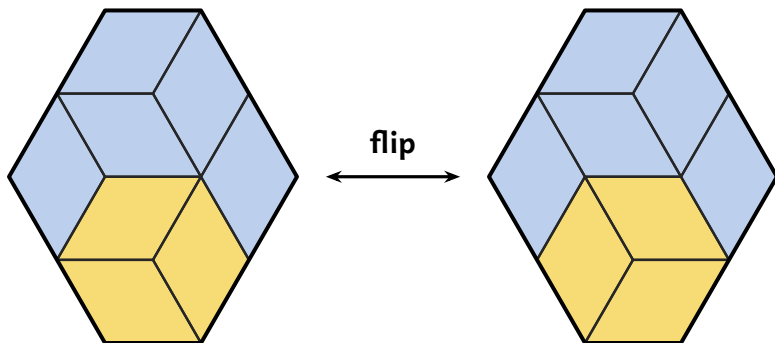
G



H

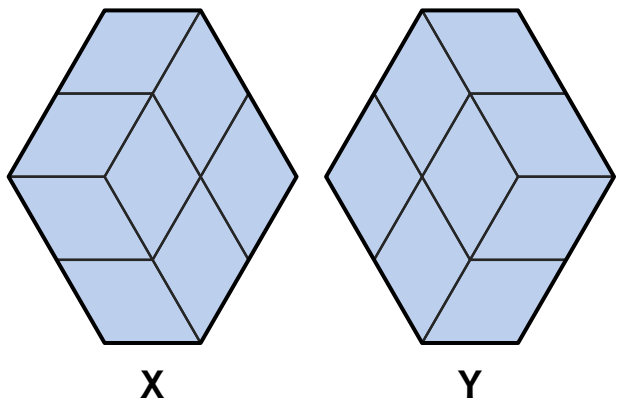
We found _____ tilings. Are you sure there are no more? Why?

3 Flips



Wherever three blues make a small hexagon (shaded yellow), you can flip them. Two tilings are **neighbors** if one flip changes one into the other.

1. Flip map. Write the letters of your tilings from page 2 in this box. Draw a line between every pair of neighbors. Move letters around until your map looks tidy.

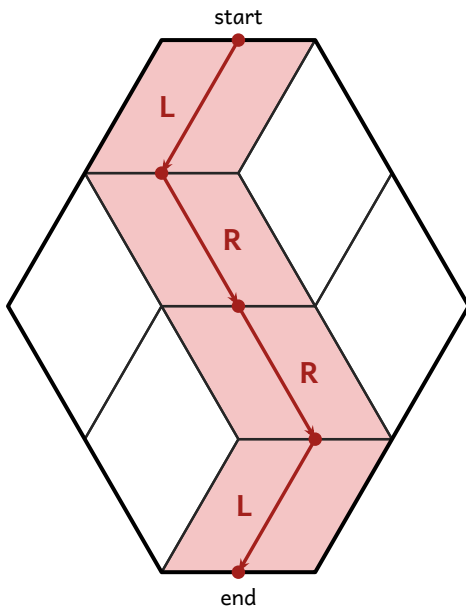


2. What is the fewest number of flips that turns X into Y? _____

How do you know it can't be done in fewer?

3. Round trips. Start at a tiling, make flips, and come back to where you started. Count the flips. Can you find a round trip with an *odd* number of flips?

4 Ribbons: a secret code



Start at the **top edge**. The blue touching it has a matching edge at its bottom. The next blue shares that edge. Keep going down to the bottom edge. This chain of blues is a **ribbon**.

Each step of a ribbon goes down-left (**L**) or down-right (**R**). The ribbon in the picture spells **L R R L L**.

1. Shade the ribbon in each of your tilings from page 2 and write its word.

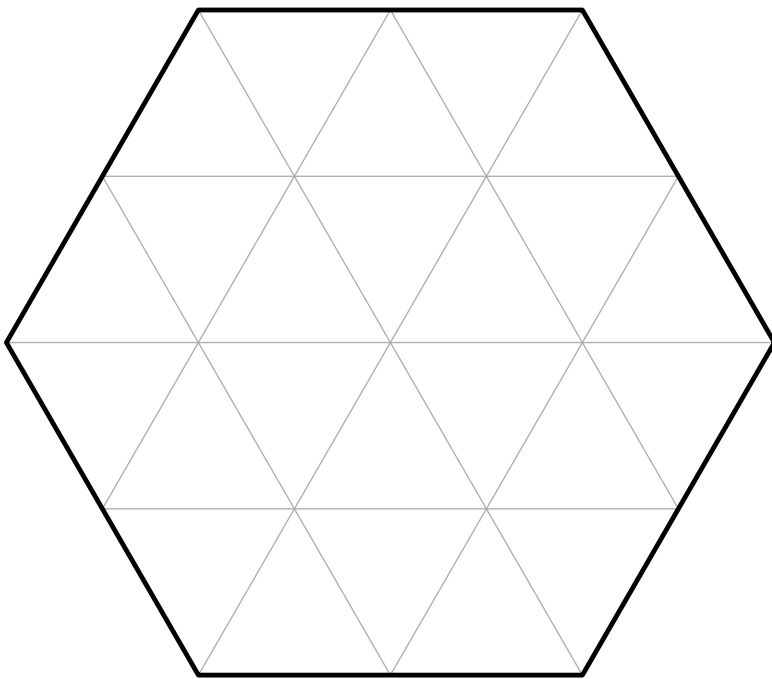
Tiling	A	B	C	D	E	F	G	H
Word								

2. List *every* word made of two L's and two R's. Does each word give a tiling? So how many tilings does the tall hexagon have?

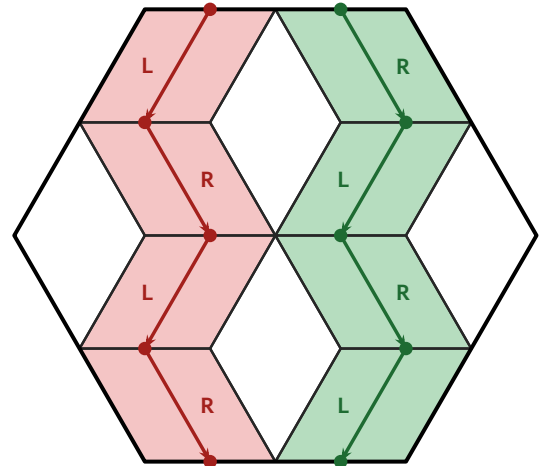
3. Pick two neighbors from your flip map and compare their words. What does one flip do to the word?

4. The **L-score** of a word is the sum of the places where the L's are. For L R R L it is $1 + 4 = 5$. What does one flip do to the L-score? Use this to explain why a round trip can *never* have an odd number of flips.

5 The big hexagon



Tile this board with 12 blues.
Now **two** ribbons start at the top edge.



Left ribbon: L R L R

Right ribbon: R L R L

1. Build a tiling and find both ribbons.

Left ribbon: _____ Right ribbon: _____

2. Can the two ribbons cross? Can they share a blue? Which pairs of words can go together? (Your flip map from page 3 can help!)

3. **Challenge.** How many tilings does the big hexagon have? Explain how you counted. Keep track on paper or a whiteboard.

4. **Look in 3D.** Shade one of your tilings like the picture. What do you see? What does a flip do to it?

